

Tenne Tian

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EDUCATION

Duke University, Pratt School of Engineering — B.S.E., Electrical & Computer Engineering (Class of 2030) 08/2026

- GPA 3.9/4.0; second major in Philosophy.

Beijing No. 80 High School — Grade 10–12

09/2024 – 07/2026

- 4.0/4.0 (Unweighted), Valedictorian

University of Pennsylvania — School of Engineering, ESAP Nanotech Program

07/2025 – 07/2025

- Completed coursework in nanotechnology (2 credits), Grade: A

SAT: 1550 (R750/M800) | Current coursework: Algorithms & Data Structures, Linear Algebra, Engineering Design, Intro to Philosophy

TECHNICAL SKILLS

Programming & ML: Python, Java, MATLAB, Swift, Excel, CAD, PCB design, ML classification pipelines (sensor-to-prediction), computer vision, statistical analysis

Hardware & Sensing: Graphene-based biosensor fabrication, ESP32 + BME688 scent sensor integration, ultrasonic/IMU sensor fusion, eye-tracking technology, LLM w/ image recognition, electrochemical workstation, SEM & AFM characterization

Systems: End-to-end sensor → app data pipeline design and deployment (4 physical pollen stations, real-time app serving 6,000+ users)

Languages: Mandarin (native), English (TOEFL: 118; R30/L30/S28/W30)

RESEARCH & TECHNICAL PROJECTS

Hardware Engineer & Drone Docking Algorithm Optimization — [Duke General Robotics Lab](#)

09/2026 – Present

- Engineer hardware for an autonomous drone-docking system, integrating drone and rover to complete Icesheet collection task.
- Optimize and test drone-docking algorithms to improve landing accuracy, reliability, and energy-efficiency.
- DIY wind tunnel by hacking fans to improve drone's wind resistance testing.

Fully Funded Researcher — Chinese Academy of Sciences (UCAS)

12/2024 – 06/2026

- Fabricated, chemically modified, and characterized graphene-based electrochemical sensors to detect pollen allergens with high sensitivity/specificity; validated results against hospital and weather-station field data.
- Authored an 18,500-word research paper (published in [Advances in Materials Physics and Chemistry](#)) and a 40,000-word experimental log documenting methodology and iteration.
- Deployed 4 Beijing monitoring stations feeding a real-time app used by 6,000+ users; the project earned a Regeneron ISEF 2026 Grand Award. [ISEF project](#)

Team Lead, ML-Based Gas Sensor — UPenn ESAP Nanotech Summer Program

07/2025 – 08/2025

- Led a 3-member team building a wearable graphene-based gas/scent detection sensor; trained a machine learning model to classify unlabeled food allergens (peanut, gluten) from raw scent-sensor data.
- Studied supporting nanotechnology fundamentals — microfluidics, lithography, band-gap theory — to inform sensor design choices feeding the ML pipeline.

Software Engineer, IRIS LLM-Powered Eye-Tracking Wearable — ENIAC XI Hackathon

06/2026

- Built a wearable that tracks eye gaze and uses an LLM to narrate what the user is looking at in real time, aimed at assisting low-vision users; owned product scoping and feature prioritization within a 3-person team.
- Designed the eye-tracking-to-LLM pipeline, translating raw gaze/image input into prompts and shaping model output for concise, real-time spoken narration.

AWARDS & HONORS

- Regeneron International Science and Engineering Fair (ISEF) 2026 **Grand Award**, Biomedical Engineering (ENBM) — \$1,000 award
- Honor Student, 2024 China National Science Talent Program; ISEF 2026 Finalist (selection rate <0.03%)
- Beijing 1st Prize + \$1,000 Scholarship + 3 Specialized Awards, China Adolescent Sci-Tech Contest (one of 2 among 718 contestants to earn both first prize and a scholarship)
- Beijing 1st Prize & National 2nd Prize, Invention Convention China — for a low-cost graphene-based pollen monitoring station (top 10%)

LEADERSHIP & COMMUNITY IMPACT

Founder, China's First School-Based Pollen Detection Station

07/2025 – 05/2026

- Built and deployed 4 monitoring stations across Beijing; launched a companion app pushing real-time data to 6,000+ users; improved measurement accuracy in wind/rain conditions and added Wi-Fi for automatic updates.

Policy Advisor, Beijing Pollen Allergy Control & Prevention

01/2026 – 03/2026

- Authored a 9,800-word policy proposal to the Beijing municipal government; contributed to passage of a new local policy regulating pollen-producing trees.